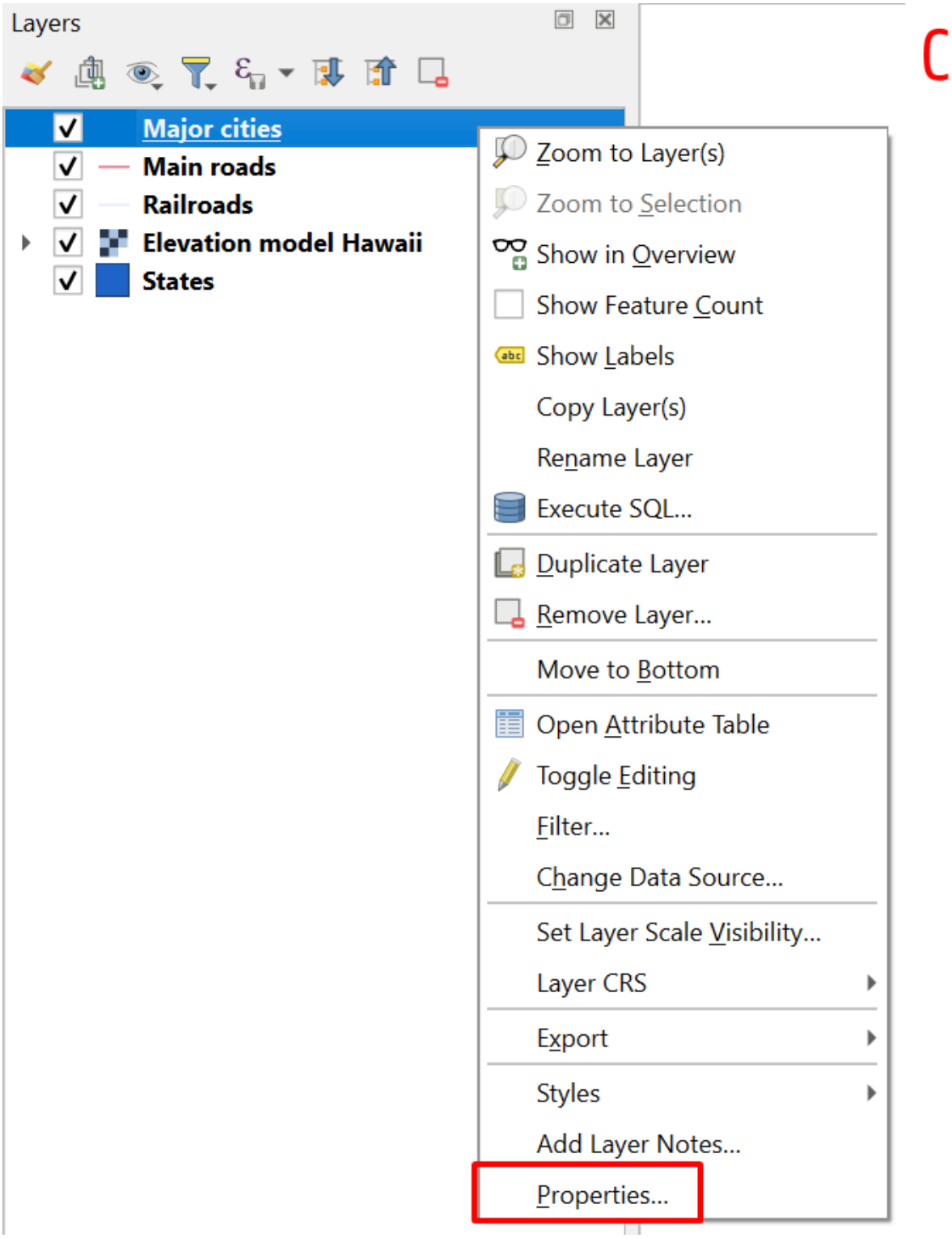
4.1 The Layer Properties window

Every layer in QGIS has its own **properties**: its name, its source file, its coordinate system, and also the way it is shown on the map (its *symbology*). This chapter is about these properties. This first page gives a short introduction: where you find these properties, and how you can look up some basic information about a layer. The pages after that deal first with vector layers (attribute table and symbology), then with raster layers (cell values and symbology).

Layer properties and actions

In The QGIS interface you saw that the *Layers* panel is one of the key parts of the QGIS interface: it gives you an overview of the layers you have loaded into your project. For each of these layers, you can carry out actions, look up information and configure settings.

If you right-click a layer, a menu opens with options and tools for that layer.



The menu includes general options, such as **renaming**, **removing**, **copying** and **duplicating** the layer, or *Zoom to Layer(s)*: the map canvas then centres and zooms so that all objects of the layer just fit in it. (As you saw on 'Navigating the map canvas', this is the same function as button A7.) Vector and raster layers also have a few options of their own.

The most important configuration window for a layer, the *Layer Properties*, can be opened with the option at the very bottom of the menu: *Properties...*.

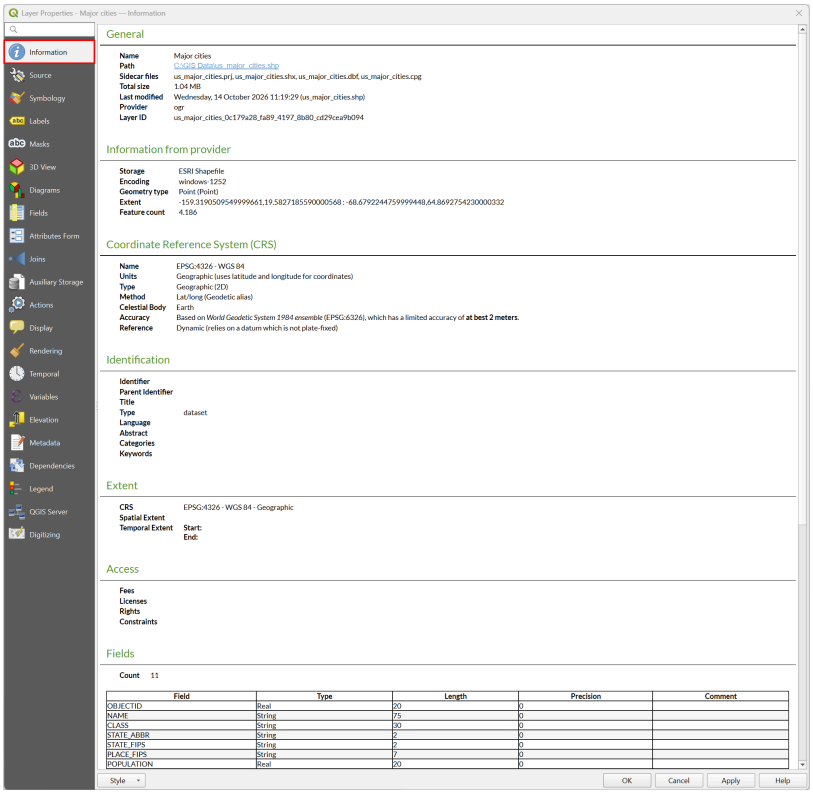
Tip

Shortcut: double-click a layer in the *Layers* panel to open its *Layer Properties*.

The *Information* and *Source* tabs

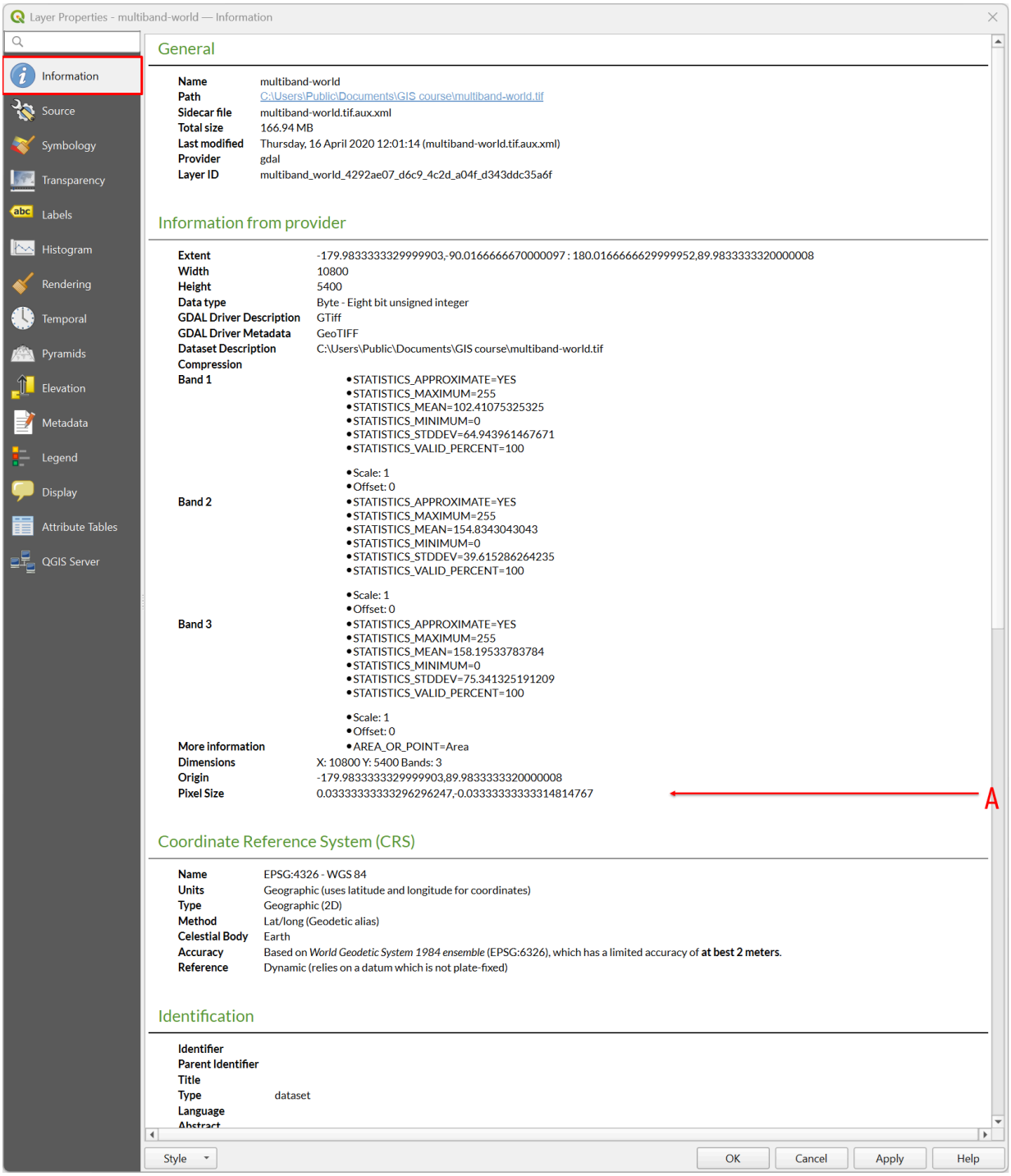
The *Layer Properties* window has partly different tabs for vector and raster layers. The first two tabs, *Information* and *Source*, are the same for both.

Information tab

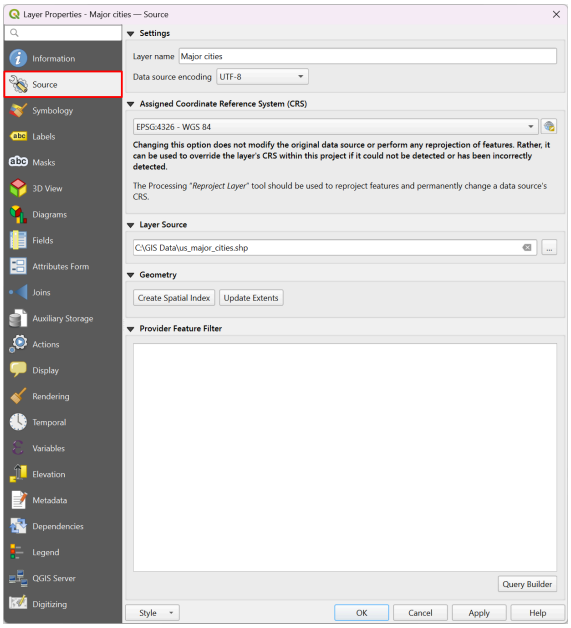


In the first tab of the *Layer Properties*, *Information*, **you will find the metadata available for the layer**: for example the file size, when the file was last modified, technical metadata such as the data type, information about licences and rights (if available)…

For vector layers this tab also gives an overview of the attribute fields and their type (for example text or number). For raster layers it shows, for example, the number of columns and rows of cells (*Width* and *Height*) and the **spatial resolution** of the raster (A: shown as *Pixel Size*, the width and height of one cell, in the units of the coordinate system, for example metres).


Layer Properties - multiband-world — Information
Information
Source
Symbology
Transparency
Labels
Histogram
Rendering
Temporal
Pyramids
Elevation
Metadata
Legend
Display
Attribute Tables
QGIS Server
General
Name multiband-world
Path C:\Users\Public\Documents\GIS course\multiband-world.tif
Sidecar file multiband-world.tif.aux.xml
Total size 166.94 MB
Last modified Thursday, 16 April 2020 12:01:14 (multiband-world.tif.aux.xml)
Provider gdal
Layer ID multiband_world_4292ae07_d6c9_4c2d_a04f_d343ddc35a6f
Information from provider
Extent -179.9833333329999903,-90.0166666670000097 : 180.0166666629999952,89.9833333320000008
Width 10800
Height 5400
Data type Byte - Eight bit unsigned integer
GDAL Driver Description GTiff
GDAL Driver Metadata GeoTIFF
Dataset Description C:\Users\Public\Documents\GIS course\multiband-world.tif
Compression
Band 1
•STATISTICS_APPROXIMATE=YES
•STATISTICS_MAXIMUM=255
•STATISTICS_MEAN=102.41075325325
•STATISTICS_MINIMUM=0
•STATISTICS_STDDEV=64.943961467671
•STATISTICS_VALID_PERCENT=100
•Scale: 1
•Offset: 0
Band 2
•STATISTICS_APPROXIMATE=YES
•STATISTICS_MAXIMUM=255
•STATISTICS_MEAN=154.8343043043
•STATISTICS_MINIMUM=0
•STATISTICS_STDDEV=39.615286264235
•STATISTICS_VALID_PERCENT=100
•Scale: 1
•Offset: 0
Band 3
•STATISTICS_APPROXIMATE=YES
•STATISTICS_MAXIMUM=255
•STATISTICS_MEAN=158.19533783784
•STATISTICS_MINIMUM=0
•STATISTICS_STDDEV=75.341325191209
•STATISTICS_VALID_PERCENT=100
•Scale: 1
•Offset: 0
More information
•AREA_OR_POINT=Area
Dimensions X: 10800 Y: 5400 Bands: 3
Origin -179.9833333329999903,89.9833333320000008
Pixel Size 0.0333333333296296247,-0.0333333333314814767
A
Coordinate Reference System (CRS)
Name EPSG:4326 - WGS 84
Units Geographic (uses latitude and longitude for coordinates)
Type Geographic (2D)
Method Lat/long (Geodetic alias)
Celestial Body Earth
Accuracy Based on World Geodetic System 1984 ensemble (EPSG:6326), which has a limited accuracy of at best 2 meters.
Reference Dynamic (relies on a datum which is not plate-fixed)
Identification
Identifier
Parent Identifier
Title
Type dataset
Language
Abstract
Style
OK
Cancel
Apply
Help

Source tab



In the second tab, *Source*, you can, among other things, change the **name** of the layer. You can also indicate which **coordinate system** and (for vector layers) which **text encoding** (for example the widely used UTF-8) should be used.

Many of the other tabs are different for vector and raster layers. The next pages therefore discuss them separately: first for vector layers, then for raster layers.